Aperture Photometry & Sky Subtraction Log

Target: Messier 82 (NGC 3034) Galactic Superwind Outflow | **Telescope/Instrument:** HST ACS/WFC | **Phase:** Phase 1 Final Extraction & Phase 2 Baseline Calibration

Coordinate Frame: ICRS / FK5	**Plate Scale:** 0.050000361 arcsec/pix	**Aperture Radius:** 25.0 arcsec (r = 0.0069444°)
Aperture Physical Radius: ~426.5 pc (at D = 3.52 Mpc)	**Aperture Area:** 1963.50 arcsec² (~785,385 pix)	**Sky Background Level (μ_{sky}):** 0.0089600 ct/pix

Aperture ID	RA (α, deg)	Dec (δ, deg)	Pixels ( )	Gross Counts ( )	Mean (, ct/pix)	Sky Pedestal ( )	Net Counts (
Sky_Bkg *(Ref)*	*148.95000*	*+69.800000*	*785,385*	*7,037.05*	*0.0089600*	*7,037.05*	*0.0*
North_outer	148.93747	+69.733217	785,384	55,813.86	0.0710657	7,037.05	48,776.8
North_mid	148.97093	+69.722209	785,388	61,801.62	0.0786893	7,037.05	54,764.5
North_inner	148.96912	+69.706520	785,385	122,879.25	0.1564570	7,037.05	115,842.2
Core (Starburst Nucleus)	**148.99183**	**+69.688112**	**785,407**	**3,006,273.60**	**3.8276600**	**7,037.05**	**2,999,236.5**
South_inner	148.97951	+69.663881	785,391	277,643.74	0.3535100	7,037.05	270,606.6
South_mid	148.99597	+69.646208	785,402	71,210.98	0.0906682	7,037.05	64,173.9
South_outer	149.04084	+69.637369	785,385	58,225.08	0.0741357	7,037.05	51,188.0

ASTROPHYSICAL REDUCTION NOTES

- **Instrumental Background Pedestal:** Derived using $C_{sky} = N_{pix} \times \mu_{sky}$. The blank sky background level of $\mu_{sky} = 0.00896\ ct\ s^{-1}\ pix^{-1}$ accounts for zodiacal scattered light, Galactic foreground, and detector readout dark bias.
- **Net Source Extraction:** Calculated via $C_{net} = C_{gross} - C_{sky}$. The starburst nucleus retains 99.77% net signal, while the faint outer plumes retain >87% net physical line and continuum emission.
- **North-South Asymmetry:** The southern inner plume ($\mu = 0.3535$) exhibits a flux enhancement factor of ~2.26× over the northern inner plume ($\mu = 0.1565$), consistent with the 80° disk inclination tilting the southern cone toward the observer.
- **Next Reduction Step:** Multiply C_{net} by HST ACS $PHOTFLAM$ and filter passband width $\Delta\lambda$ to obtain calibrated integrated band fluxes ($erg\ s^{-1}\ cm^{-2}$).